

JESSICA CHEN

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PROFESSIONAL EXPERIENCE

Technical Design Intern

May 2026 – August 2026

Sledgehammer Games (Activision)

Toronto, Canada

- Developed gameplay concepts and prototypes for a future **Call of Duty** release using proprietary tools, translating design ideas into playable experiences for evaluation
- Built **modular, player-facing gameplay** ingredients to support flexible and reusable gameplay experiences
- Iterated on design solutions with designers and producers, applying playtesting feedback and team reviews

PROJECTS

Fitting In | *Unity, C#*

- Collaborate with a group of programmers, audio designers, and artists to develop a **local multiplayer game** where players collaboratively control a human-like spaceship to complete everyday tasks
- Engineered the spaceship's walking system to translate distributed player inputs into coordinated locomotion, handling directional movement, gravity, ground detection, and movement feedback
- Built **interactive gameplay tasks**, combining physics-based object manipulation, collision detection, state-driven logic, and player feedback to create responsive task interactions

Paper Plate Paranoia | *Unity, C#*

- Designed a **bullet hell game** with a group of 4 using a paper-plate character concept and core gameplay loop of collecting items while surviving enemy attack patterns
- Programmed enemies including a ketchup laser and three-stage exploding popcorn, implementing attack behaviors, gameplay states, and **Unity Animator Controllers** for synchronized animation transitions
- Designed dynamic enemy attack patterns, combining tracking behaviors and telegraphed area attacks to create varied, readable combat encounters

Heimer Academy (1st Place Rift Rewind Hackathon 2025) | *Python, AWS*

- Built a personalized **AI coaching agent** that analyzes League of Legends players' in-game performance and champion data to provide actionable improvement insights and recommendations
- Developed a **graph-based champion recommender**, learning champion associations from players' historical selections and combining them with archetype and playstyle similarity from **Riot Data Dragon**
- Integrated the recommender with the champion ability-analysis backend, dynamically reweighting recommendations based on recent player performance and filtering out recently played champions

ADDITIONAL WORK

UofT eSports / Alberta Esports Association Shoutcaster

March 2021 – May 2024

- Coordinated with co-casters and production teams to deliver live commentary and strategic analysis to 200+ viewers
- Analyzed live League of Legends game states, tracking player positioning, resources, objective control, and team strategies to anticipate plays and explain tactical decisions

Quantitative Trading Analyst / AI Engineer

August 2024 – August 2025

RBC Capital Markets

Toronto, Canada

- Developed low-latency algorithm that optimizes 150+ automated strategies through modular signal processing
- Consolidated 8+ predictive models and 5+ years of multi-exchange data into unified **PyTorch** training pipelines for large-scale analysis and model development

Undergraduate Research Assistant | Supervisor: Dr. Michael Bowling

University of Alberta, Department of Computing Science

Edmonton, Canada

- Co-authored** a paper accepted to the **Reinforcement Learning Conference 2026**, introducing a reinforcement learning framework that allows agents to explicitly control their computational usage
- Extended DQN with temporally extended actions, reducing inference calls by 75% while maintaining reward performance

TECHNICAL SKILLS

Languages: Python, C/C#/C++, JavaScript, TypeScript, Java, GSC, SQL, Q, GDScript

Frameworks/Libraries: React, PyTorch, Jax, TensorFlow, Scikit-learn, Gymnasium, Numpy, Matplotlib, Pandas

Developer Tools: Git, Unity, Godot, Microsoft Azure, Figma, WandB, Supabase, Firebase, Django

EDUCATION

University of Toronto | cGPA 3.92/4.00

Toronto, ON

Bachelor of Science in Computer Science, Statistical Science, Mathematics

Graduated 2026